Lossy Image Compression with Conditional Diffusion Models: Implementation & Evaluation (NeurIPS 2023)

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Date : 2025.12.15

Outline

- Paper Overview
- Generative Diversity & Stability Analysis of CDC
- Domain Shift Evaluation
- Effect of conditioning scale

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Paper Overview - Introduction

- VAE-based Codec
  - Strong rate–distortion performance, optimized for MSE distortion.
  - Gaussian decoder → mode averaging → blurry reconstructions.
  - Limited texture expressiveness at low bitrates.

- GAN-based Codec
  - Great perceptual realism, especially at low bitrates.
  - But unstable training + GAN artifacts.
  - Hard to tune; distortion–perception balance difficult to control.

- Traditional Codec
  - Hand-crafted transforms (JPEG/BPG/WEBP); fast and efficient.
  - Limited modeling capacity → artifacts at low bitrates.
  - Relatively poor perceptual quality vs. learned codecs; non–data-driven.

Paper Overview - Introduction

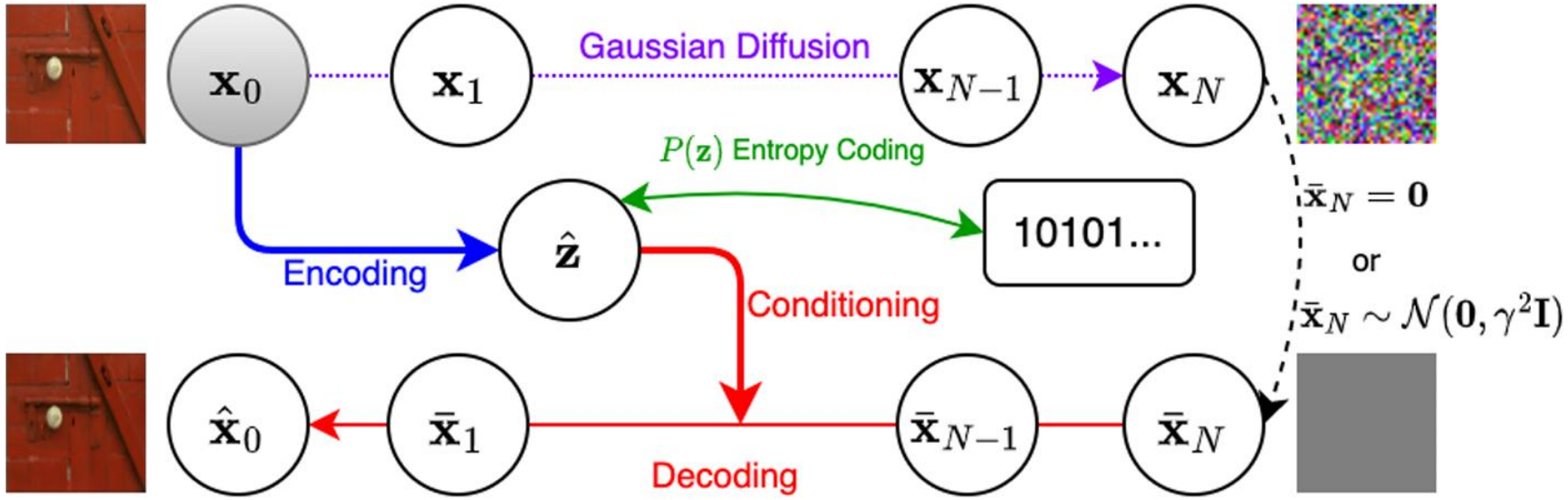
- CDC (Conditional Diffusion Compression)
  - Transform coding maps images to latent space for entropy coding.
  - CDC replaces the Gaussian decoder with a conditional diffusion model.
  - Diffusion provides higher expressiveness → improved perceptual quality.

- Stochastic Decoder vs. Deterministic Decoder
  - **Content variable ($\hat{z}$):** compressed semantic info. (compressed and transmitted!)
  - **Texture variables ($x_1 \ldots x_N$):** high-frequency details generated at decode time. (don't get transmitted)
  - **Deterministic decoding ($\gamma = 0$):** smooth, lower distortion.
  - **Stochastic decoding ($\gamma > 0$):** richer texture, better perceptual quality.
  - Tradeoff between distortion and perceptual quality.

- Parameterization-X-Prediction
  - Allows high-quality reconstructions in only a handful of steps (17 vs. 1000), practical for real-world use.

Paper Overview - Method

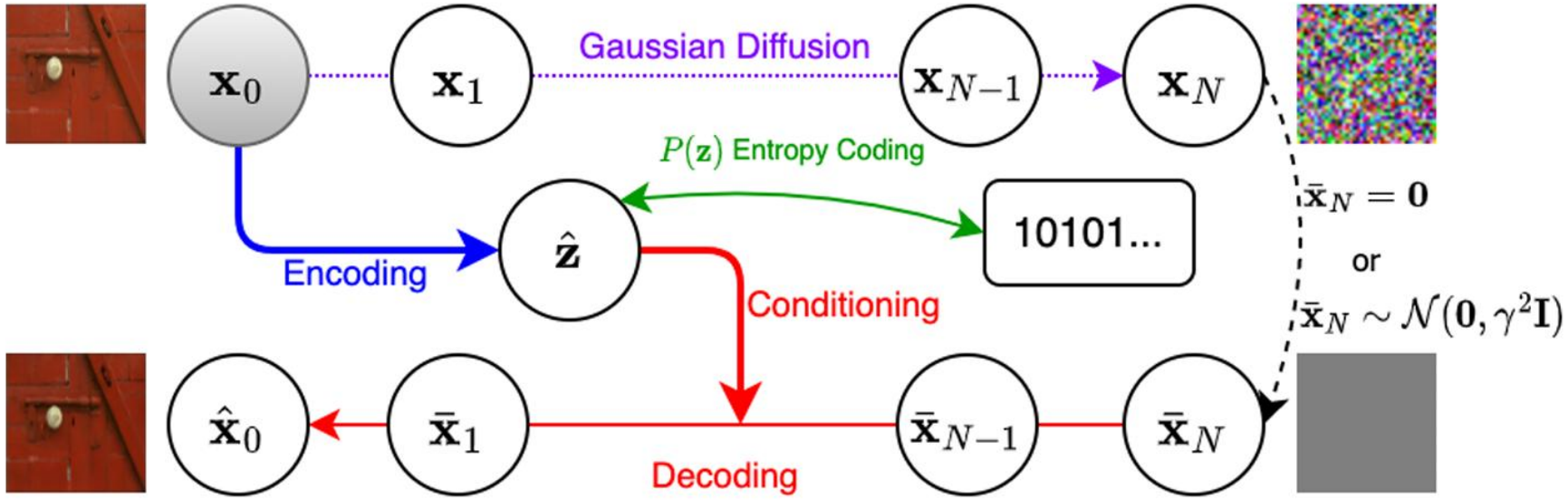
- Decoding Process
  - Entropy-decode z using the prior p(z) and conditionally decode the image x_0 using ancestral sampling.
  - Consider two decoding schemes:
    - Deterministic: $x_N = 0$ (or $\gamma = 0$) → stable, smooth reconstructions
    - Stochastic: $x_N \sim N(0,\gamma 2I)$ (where $\gamma > 0$) → richer textures, more perceptual detail
- "Texture" Variables
  - Since the variables $x_{1:N}$ are not stored but generated at test time, these "texture" variables can result in variable reconstructions (slightly different but perceptually similar) upon stochastic decoding.



Paper Overview - Method

(a) A deterministic decoding scheme

(b) A stochastic decoding scheme



Paper Overview - Method

- Fast Decoding using X-prediction
  - Iterative generative process is a major roadblock towards fast generation.
  - Other methods require additional post-processing methods, such as progress distillation and parallel denoising.
  - X-prediction parameterization behaves like an autoencoder at low noise levels, enabling fast reconstruction.
    - X-prediction model with only 17 decoding steps achieves comparable performance to the ϵ-model with hundreds of steps.
    - Without the need of any post-processing.

- Optional Perceptual Loss
  - Influence the perceptual quality of the compressed images by introducing additional loss functions (**LPIPS**).
  - This loss term is weighted by an additional Lagrange multiplier $\rho \in [0,1)$, resulting in a three-way tradeoff between rate, distortion, and perceptual quality.
  - It is proven that mathematically that it is impossible to simultaneously obtain arbitrarily good perceptual qualities and low distortions.
    - In this paper, we observe a similar fundamental tradeoff between perception and distortion.

Paper Overview - Experiments

- Metrics
  - Perceptual metrics: FID, LPIPS, PIEAPP, DISTS…
  - Distortion metrics: FSIM, MS-SSIM, SSIM, PSNR…

- Test Data
  - Kodak
  - Tecnick
  - DIV2K
  - COCO2017

- Model Training
  - Vimeo-90k dataset

Paper Overview - Experiments

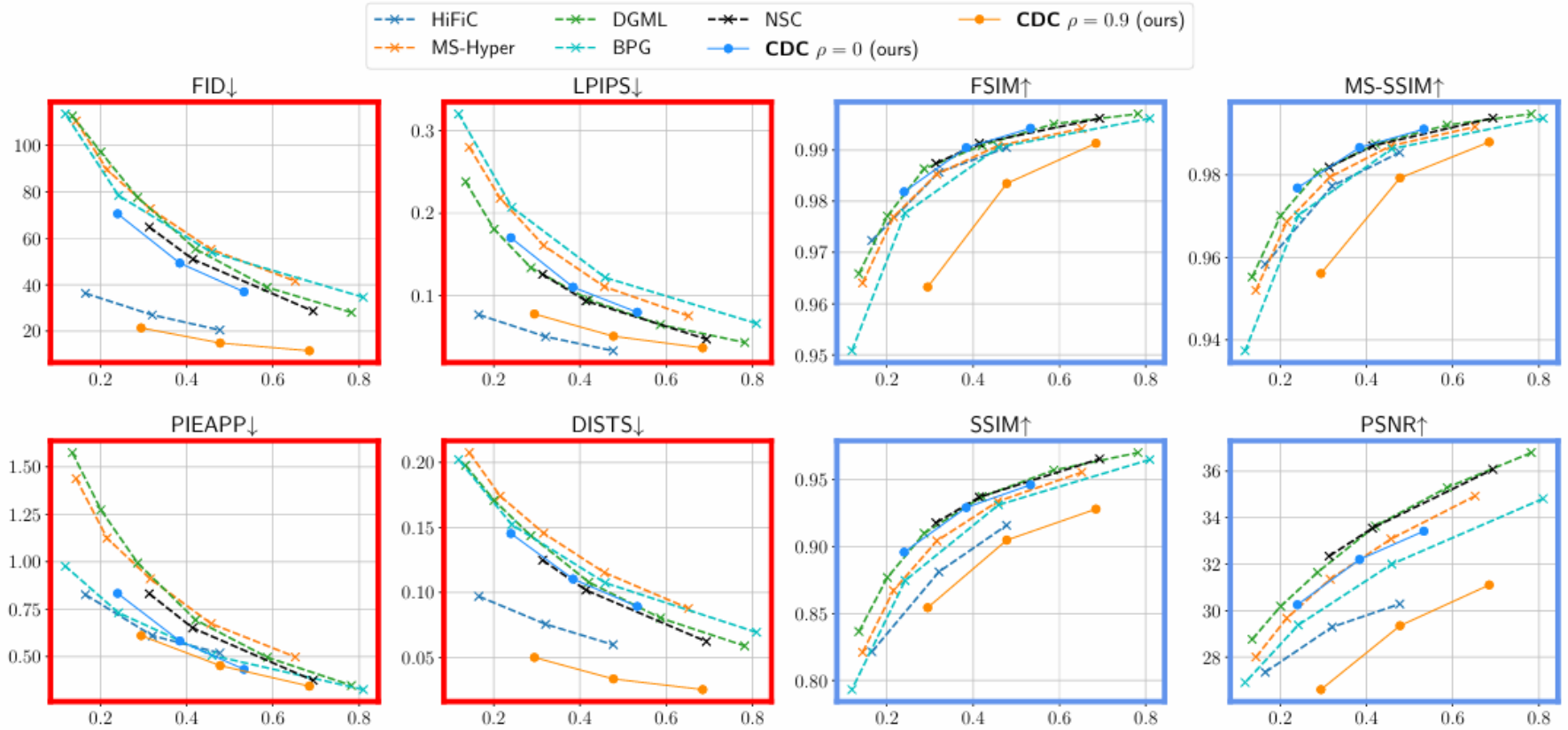


Figure 2: Tradeoffs between bitrate (x-axes, in bpp) and different metrics (y-axes) for various models tested on DIV2K. We consider both perceptual (red frames) and distortion metrics (blue frames). Arrows in the plot titles indicate whether high (↑) or low (↓) values indicate a better score. CDC (proposed) in its basic version (deterministic, without finetuning to LPIPS) compares favorably in distortion metrics, while CDC with stochastic decoding and added LPIPS losses performs favorably on perceptual metrics.

Paper Overview - Experiments

(a) Ground Truth (b) CDC($\rho = 0.9$) (bpp=0.205) (c) HiFiC (bpp=0.207)

(d) Ground Truth (e) CDC($\rho = 0.9$) **(bpp=0.398)** (f) HiFiC (bpp=0.456)

Figure 3: Reconstructed Kodak images (cropped images, see full images in Appendix F). 1st row: compared to HiFiC under similar bitrate, our model retains more details around the eyes of the parrot. 2nd row: our model still gets slightly better visual reconstruction than HiFiC while using *less* bitrate.

Paper Overview - Experiments

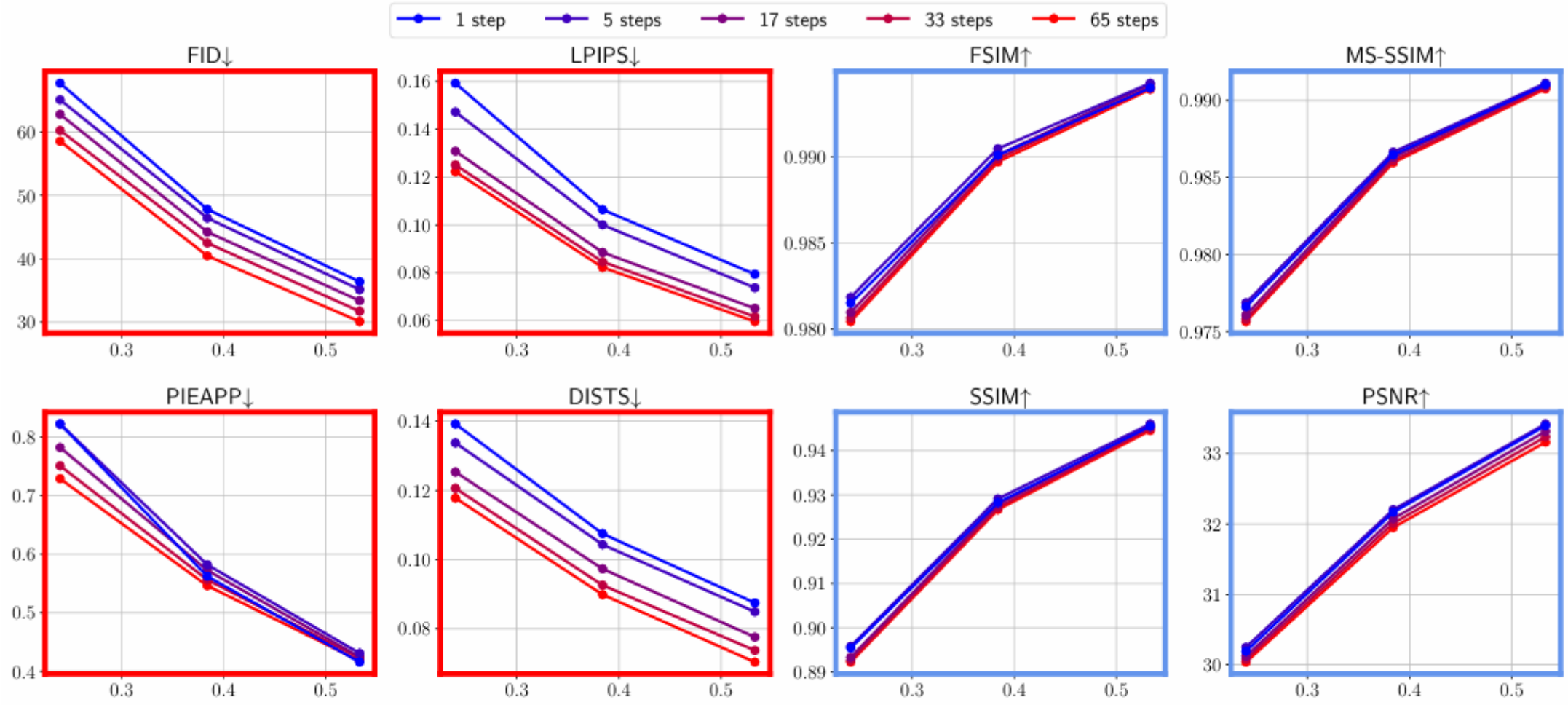


Figure 4: Compression performance with different numbers of decoding step. We use $\gamma = 0$ (deterministic decoding) to plot distortion curves and $\gamma = 1$ for perceptual quality curves.

Paper Overview - Experiments

(a) Ground Truth

(b) $\gamma = 0$
MS-SSIM=0.979
LPIPS=0.078

(c) $\gamma = 0.6$
MS-SSIM=0.977
LPIPS=0.064

(d) $\gamma = 0.8$
MS-SSIM=0.973
LPIPS=0.054

(e) $\gamma = 1$
MS-SSIM=0.964
LPIPS=0.058

Figure 5: Qualitative comparison of deterministic and stochastic decoding methods. Deterministic decoding typically results in a smoother image reconstruction. By increasing the noise γ used upon decoding the images, we observe more and more detail and rugged texture on the face of the sculpture. ($\gamma = 0.8$) show the best agreement with the ground truth image. All the images share the same bpp.

Paper Overview - Limitations

- Potential Hallucination
  - Stochastic decoding may generate **hallucinated textures**.
  - At very low bitrates, generated details may **misrepresent true image content**.

- Identity / Semantic Risks
  - When compressing human faces or critical details, hallucinated textures may **alter identity or semantics**, raising concerns for fairness & trustworthiness.

Figure 14: We stochastically decode the same latent variable z and $\gamma = 0.8$ but different random seed for $x_N \sim N(0,\gamma 2I)$. Different random seeds may yield low-level textural distinction.

Outline

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- Generative Diversity & Stability Analysis of CDC
- Domain Shift Evaluation
- Effect of conditioning scale

Generative Diversity & Stability Analysis of CDC - Motivation

- Goal: analyze the **stochastic decoding behavior** of CDC under $\gamma > 0$.
  - Stochastic decoding introduces **non-deterministic texture generation**, while content remains unchanged.

- CDC separates information into:
  - **Content latent ($\hat{z}$):** preserves global semantics
  - **Texture variables ($x_1 \ldots x_N$):** synthesized during decoding

- Key hypothesis:
  - Randomness should only affect **high-frequency textures.**
  - Global structure and semantics should remain **stable and consistent** across samples.

- Purpose: verify whether CDC maintains **semantic stability** while providing **diverse but controlled textures**.

Generative Diversity & Stability Analysis of CDC - Setup

- **Model Architecture:**
  - CDC X-parameterization (LPIPS optimized).
- **Test Data:**
  - Kodak Dataset (kodim02.png).
- **Parameters:**
  - Stochastic noise intensity gamma = **0.8**.
- **Environment:**
  - Inference performed on CPU.

Generative Diversity & Stability Analysis of CDC - Experiment A

- **Experiment A: Residual & Consistency Analysis**
  - **Method:** We fixed the latent variable **z** and performed decoding twice using different random seeds to analyze the pixel-wise differences.

- **Quantitative Metric:**
  - **PSNR: 35.10 dB**

- **Qualitative Observation:**
  - The **Residual Heatmap** shows that variations are concentrated on object edges and surface materials (e.g., wood grain, rust).
  - Smooth regions (e.g., red paint) exhibit minimal residual variance.

Generative Diversity & Stability Analysis of CDC - Experiment A

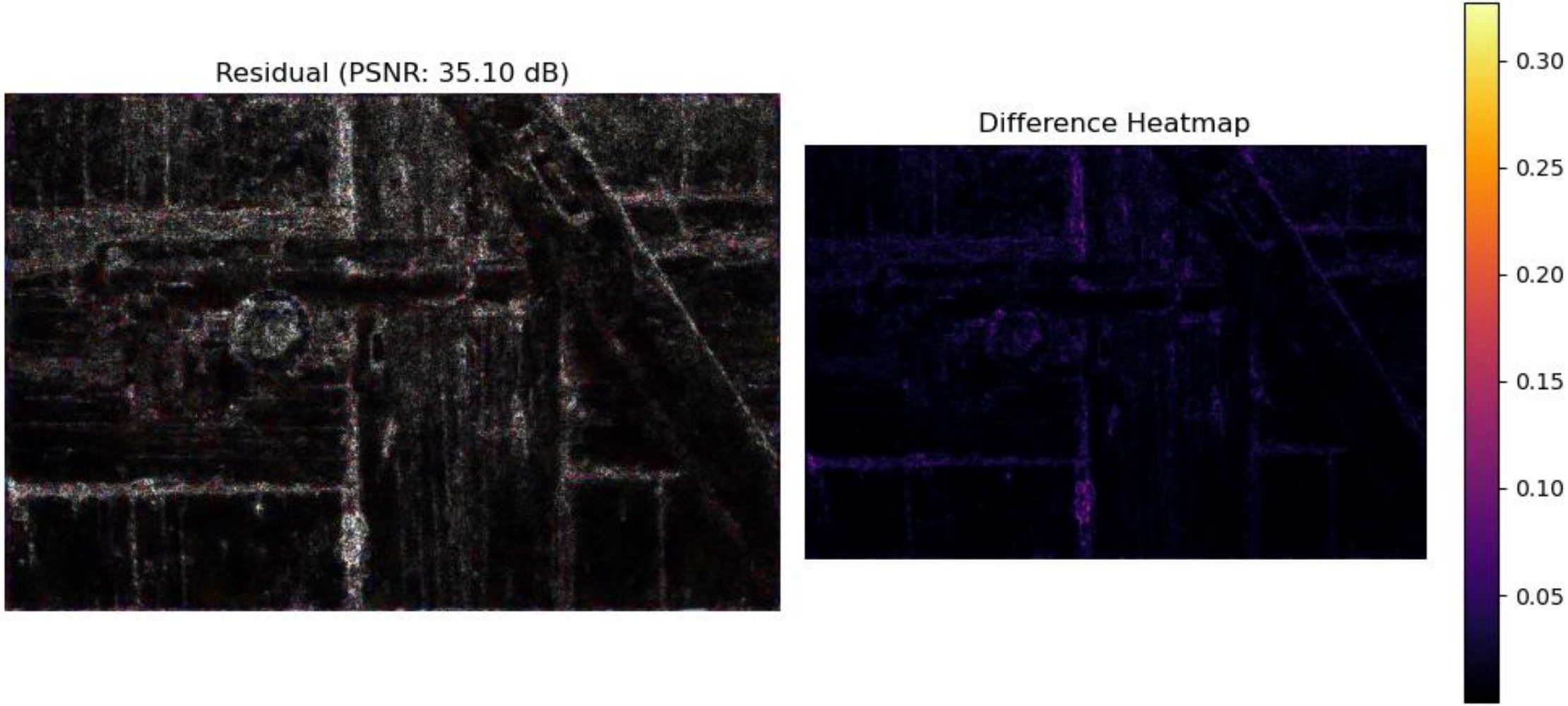
- **Experiment A: Residual & Consistency Analysis**
  - **Stability test: Gamma = 0.8**

Decode 1 (Seed 100)

Decode 2 (Seed 200)

Generative Diversity & Stability Analysis of CDC - Experiment A

- **Experiment A: Residual & Consistency Analysis**
  - **Stability test: Gamma = 0.8**



Generative Diversity & Stability Analysis of CDC - Experiment B

- **Experiment B: Frequency Domain Stability (Blur Test)**
  - **Method:** We generated 10 stochastic samples (N=10) for the same compressed image, calculated the pixel-wise variance, and compared the "Original Variance" against the "Gaussian Blurred Variance" (Low-pass filtered).

- **Data:**
  - Original Average Variance: 0.000153
  - Blurred Average Variance: 0.000006
  - **Variance Drop Rate: 96.2%**

Generative Diversity & Stability Analysis of CDC - Experiment B

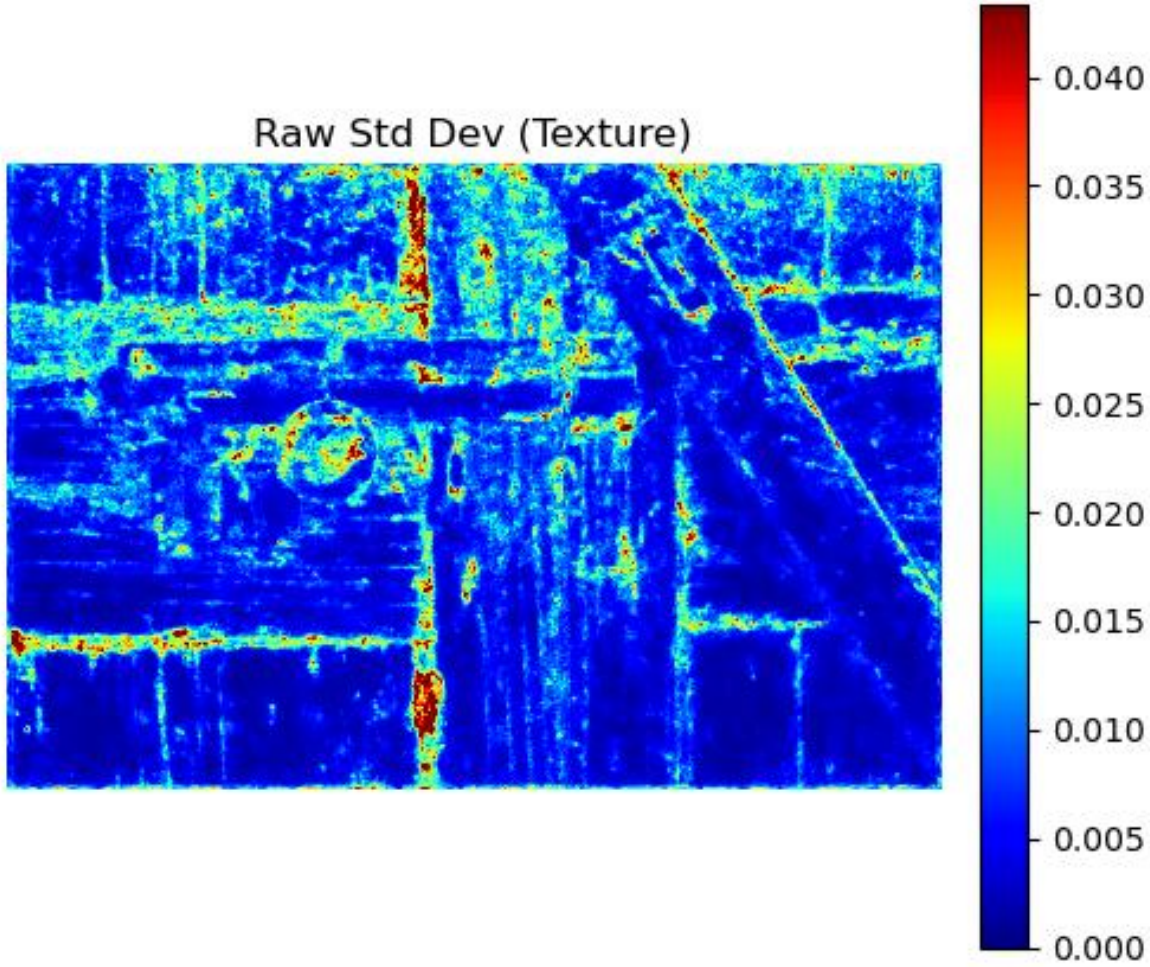
- **Experiment B: Frequency Domain Stability (Blur Test)**
  - Blur test: Variance drop = 96.2%

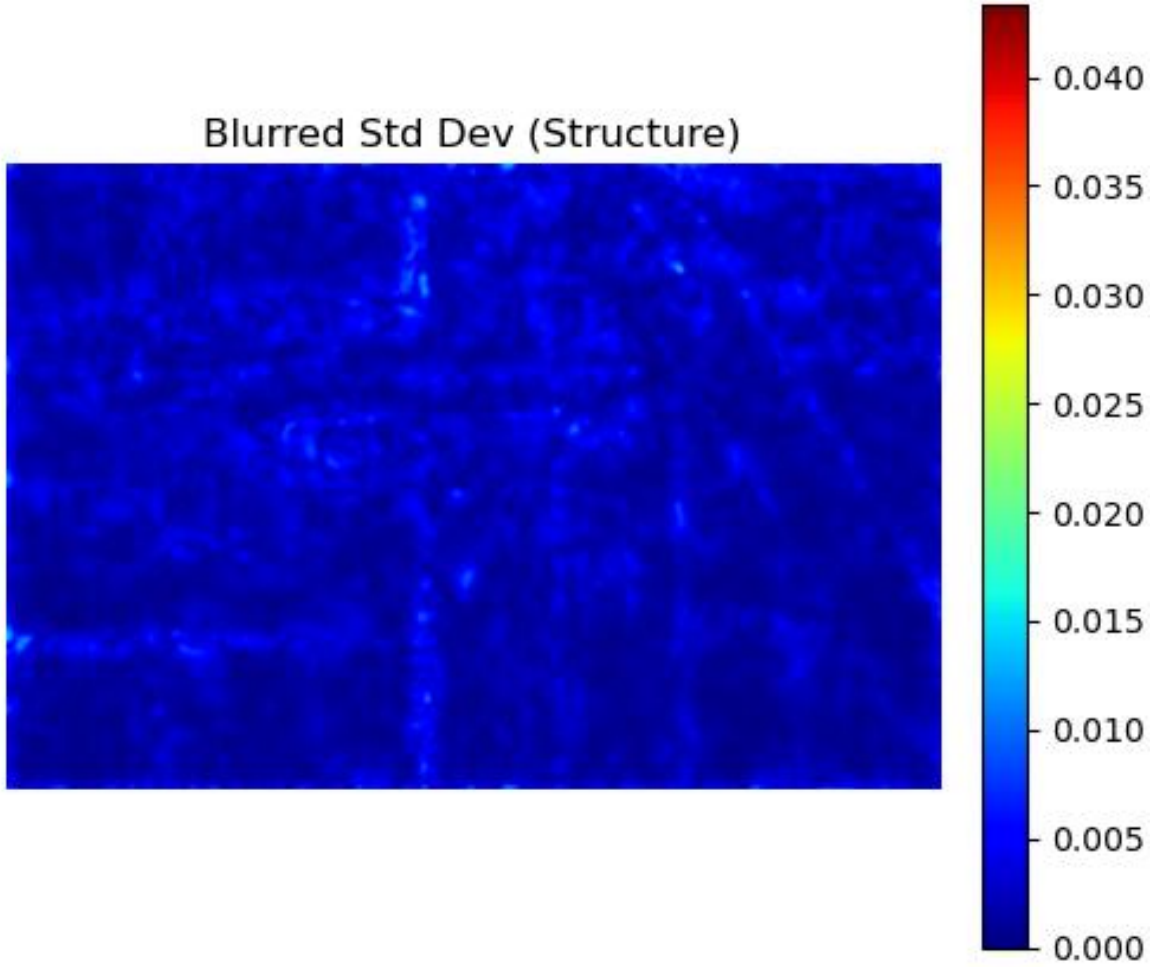
Sample Decoded Image

Blurred Image (Structure Only)

Generative Diversity & Stability Analysis of CDC - Experiment B

- **Experiment B: Frequency Domain Stability (Blur Test)**
  - Blur test: Variance drop = 96.2%





Generative Diversity & Stability Analysis of CDC - Discussion

- Structural Stability
  - Experiment A produced **35.10 dB PSNR**, well above the 30 dB high-quality threshold.
  - Confirms stochastic decoding introduces differences, but **changes are small and non-structural**.
  - Demonstrates CDC maintains **stable global semantics** across multiple decoding variations.

- Frequency Characteristics of Texture Synthesis
  - Experiment B shows a **96.2% variance drop** after removing high-frequency components.
  - Indicates that **over 96% of stochastic variation lies in high-frequency textures** (noise, fine detail).
  - Low-frequency structure remains **consistent across 10 stochastic decoding variations**.
  - Strong statistical evidence that randomness is **localized to perceptual texture regions**.

- Visual Verification
  - Visual comparison confirms variations occur **exactly in texture regions** (e.g., wood grain, metal rust).
  - No random artifacts or semantic distortions observed.
  - Shows CDC performs **semantically aligned texture synthesis**, not arbitrary noise generation.

Generative Diversity & Stability Analysis of CDC - Conclusion

- CDC achieves the core goal of perceptual compression:
  - **stable content + stochastic high-frequency texture synthesis**.

- The **96.2% variance drop rate** quantitatively demonstrates that randomness is confined to detail layers.

- CDC avoids the **blurriness of traditional VAEs** by generating realistic textures where perceptual sensitivity is low.

- Confirms CDC's decoding process is **diverse, controlled, and semantically stable**.

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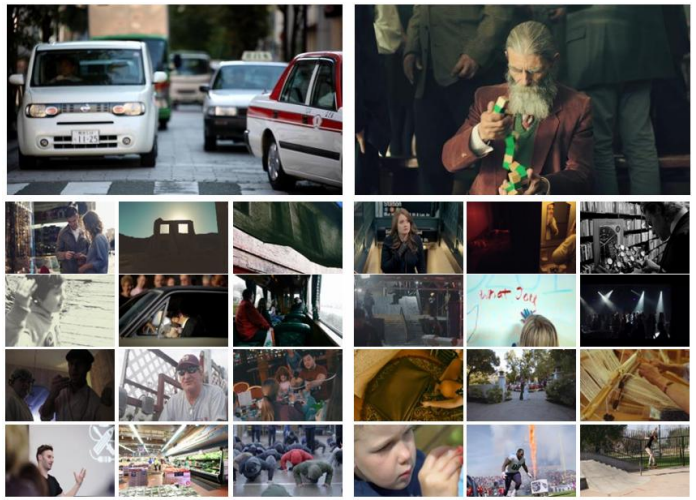
Domain Shift Evaluation

- Objective:
  - CDC is trained on natural videos (Vimeo-90k). How does it handle non-natural line art (Manga109)?
  - Evaluate the CDC model on Out-of-Distribution (OOD) data.
- Key Findings:
  - Traditional codecs (BPG) significantly outperform CDC in PSNR on line art.
  - stochastic decoding strategy ($\gamma = 0.8$) yields uncertain effects on compression image quality.
- Conclusion:
  - Diffusion models require domain adaptation; blind application to OOD data causes artifact generation.



Vimeo-90k

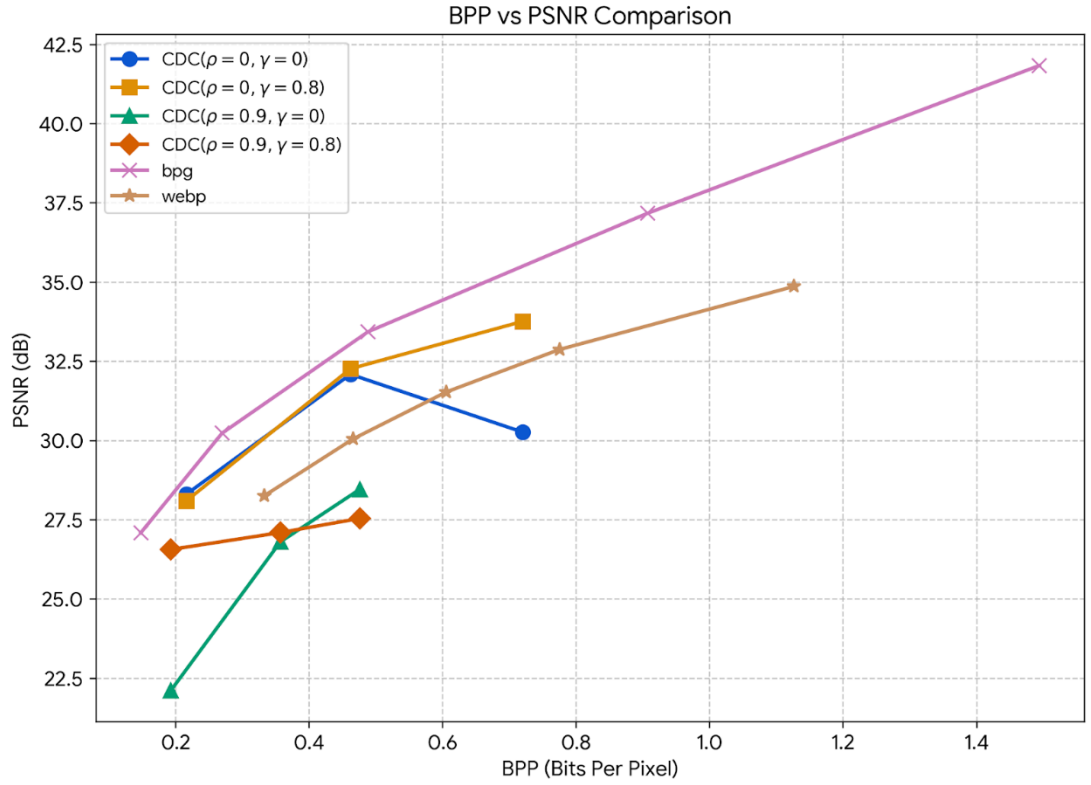
Manga109

Experimental Setup

- Dataset (OOD):
  - **Manga109**: High-contrast line art, flat colors, screen tones.
  - Contrast: Lack of stochastic high-frequency noise found in natural photos.
- Baselines (Traditional Codecs):
  - **BPG (HEVC):** State-of-the-art benchmark for PSNR.
  - **WebP:** Standard web codec.
- CDC Model Variants:
  - **$\rho = 0$ (MSE-optimized):** Deterministic decoding ($\gamma = 0$).
  - **$\rho = 0.9$ (Perceptual-optimized):** Tested with Deterministic ($\gamma = 0$) and Stochastic ($\gamma = 0.8$) decoding.
- Evaluation Metric:
  - PSNR

Overall Results

- Observation:
  - CDC fails to match BPG on the OOD dataset
  - The perceptual model (ρ = 0.9) suffers a catastrophic drop in fidelity on line art.

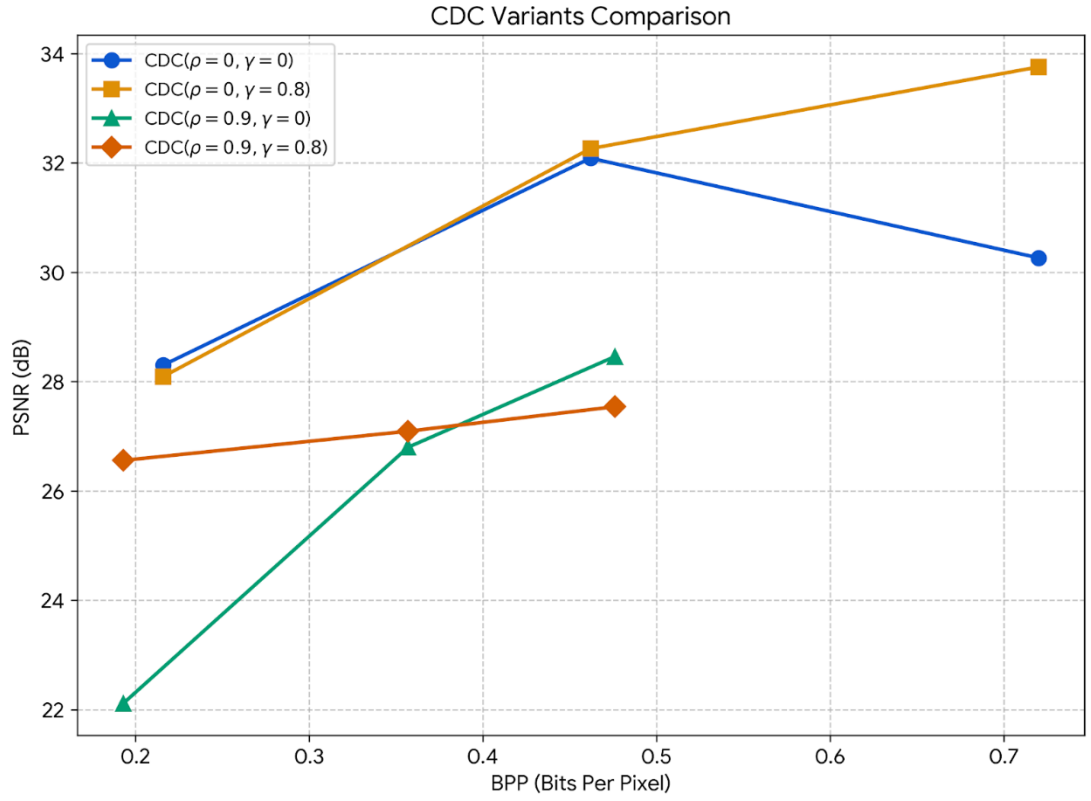


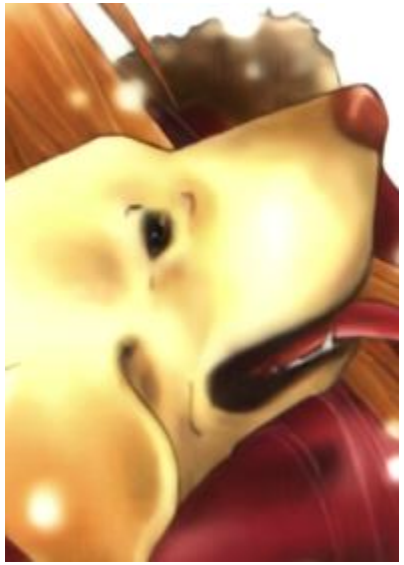
(a) CDC(ρ=0.9,γ=0.8) (b) BPG (c) WebP (d) Original image

Qualitative comparison of reconstructed image

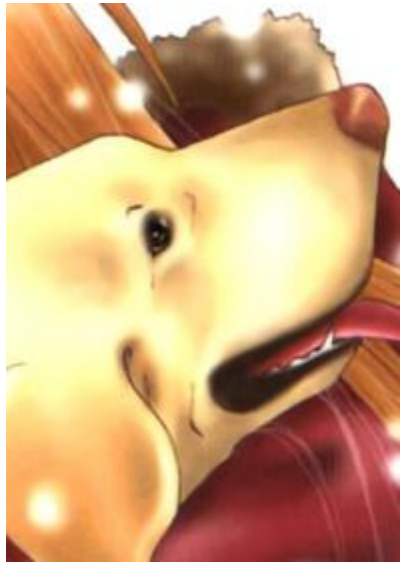
Optimization Comparison

- Observation:
  - MSE-optimized (ρ = 0) model performs better than Perceptual-optimized (ρ = 0.9) model.
    - The perceptual "details" are effectively noise, not useful texture.





(a) ρ = 0.9

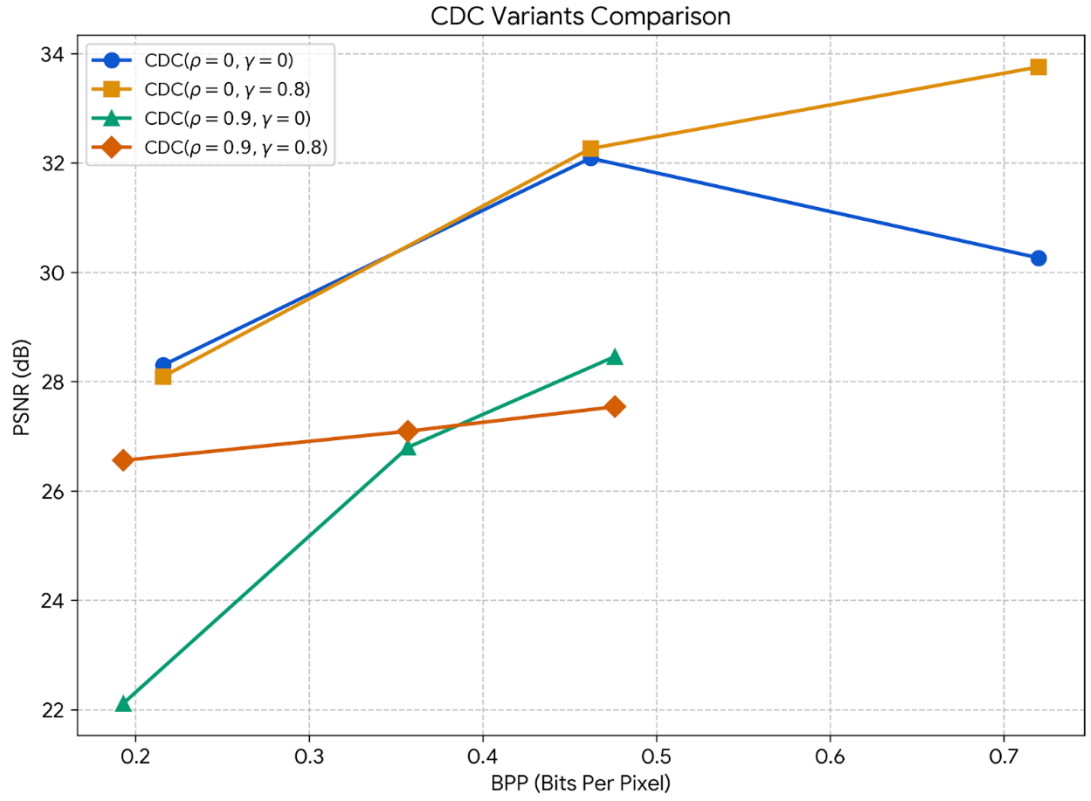


(b) ρ = 0

(c) original

Impact of Stochastic Decoding

- Observation:
  - The stochastic decoding strategy ($\gamma = 0.8$) yields uncertain effects on compression performance.
    - The MSE-optimized model performs better at high BPP, while the perceptual-optimized model is more effective at low BPP.



(a) $\rho = 0$, $\gamma = 0.8$
high BPP

(b) $\rho = 0.9$, $\gamma = 0.8$
low BPP

(c) original

Discussion & Conclusion

- Texture Hallucination:
  - The model learns that “real images have noise”, and forces this learned noise onto clean manga surfaces.
  - Result: Grainy artifacts instead of clean lines.
- The Domain Gap:
  - BPG uses predictive coding → Domain Agnostic → Works well on edges.
  - CDC is Data-Driven → Domain Specific → Fails when Training ≠ Testing.
- Conclusion:
  - **Generalization Limit:** Untuned diffusion models cannot generalize well to non-natural image domains.
  - **Traditional Codecs Win:** BPG remains the superior choice for line art compression (high fidelity, no hallucinations).
  - **Risk:** "Perceptual quality" metrics (like FID) on OOD data can be misleading if the generated texture is semantically incorrect.

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Evaluation – Effect of conditioning scale

- Denoising process in CDC
  - Restore image from given semantic info.
  - Input latent (from encoder) is served as conditioning in denoising diffusion
  - Heuristic: Does stronger or weaker input latent improve the result?

$$p_\theta(\mathbf{x}_{0:T}|\mathbf{z}) = p(\mathbf{x}_N)\prod p_\theta(\mathbf{x}_{n-1}|\mathbf{x}_n,\mathbf{z}) = p(\mathbf{x}_N)\prod \mathcal{N}(\mathbf{x}_{n-1}|M_\theta(\mathbf{x}_n,\mathbf{z},n),\beta_n\mathbf{I}).$$

- Guidance scale
  - From Classifier-Free Guidance
  - The larger of s (scaling factor), the denoising process follow condition more
  - 0 means totally unconditional
  - We only take the concept of scaling factor to control the strength of i/p latent

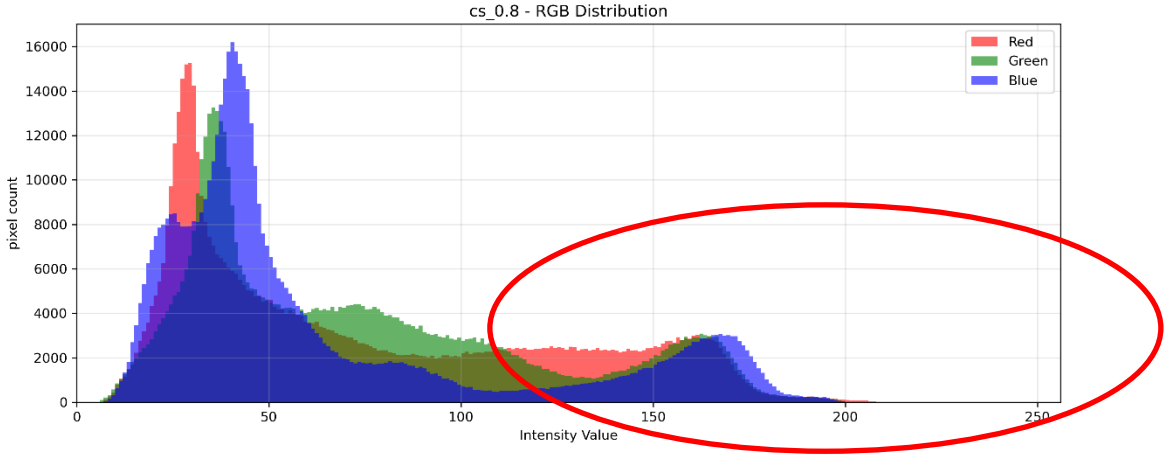
$$\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t|y) = \underbrace{\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t)}_{\text{unconditional score}} + \underbrace{s(\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t|y) - \nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t))}_{\text{unconditional to conditional}}$$

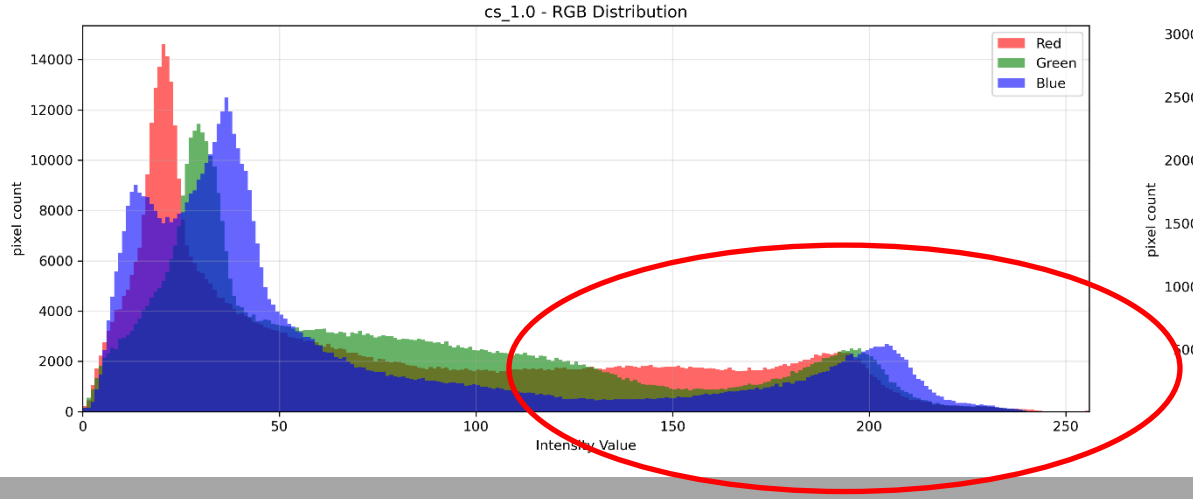
Evaluation – Effect of conditioning scale

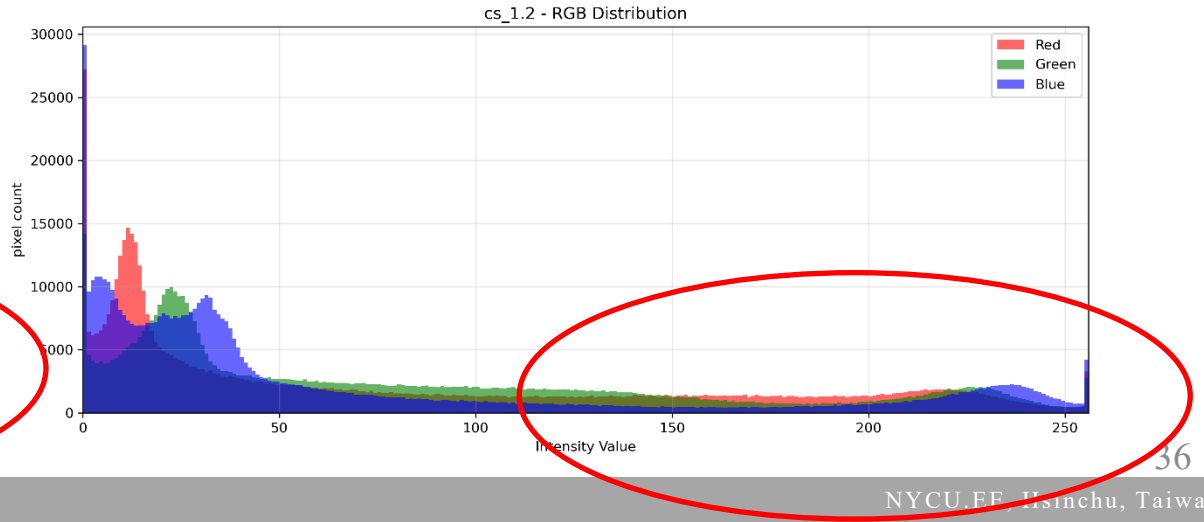
- Our test
  - Scaling the input latent (from encoder)
  - Sweep scale from 0.9 to 1.1, stride = 0.1+ scale = 0.8, 0.85, 1.15, 1.2
  - $\gamma = 0, 0.8$
  - Using best setting mentioned on paper: Add perceptual loss (Lagrange multiplier = 0.9), denoising step = 17
- Metrics
  - Perceptual metrics: FID, LPIPS
  - Distortion metrics: SSIM, PSNR
  - We take average value of LPIPS, PSNR, SSIM across 25 images
- Test Data
  - DIV2K
  - Resize the images with the shorter dimension being equal to 768px. Then, each image is center-cropped to a 768×768 squared shape

Evaluation result (γ = 0.8)

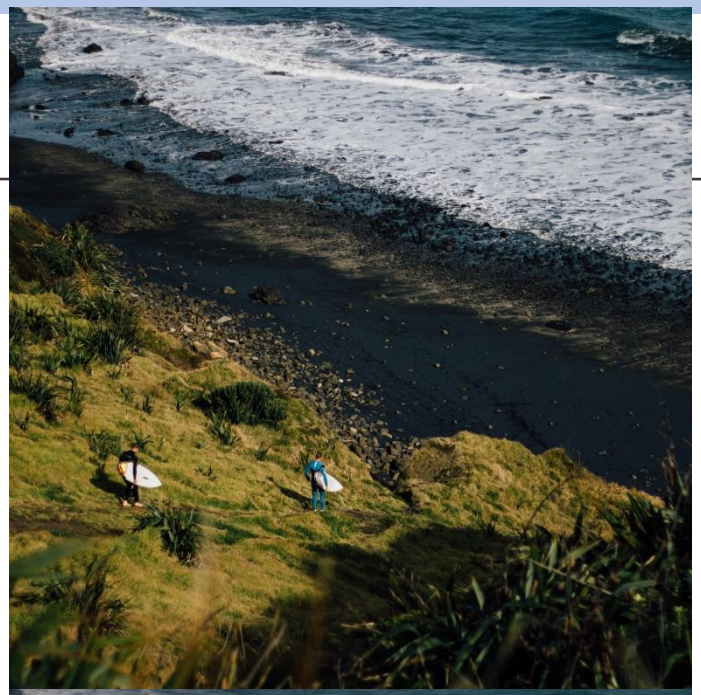
- Effect on compressed image
  - If cond_scale increase, then pixel intensity will also increase. Brightness, sharpness and contrast increase
  - If decrease, pixel intensity (contrast) will decrease. Image become blur and less contrast



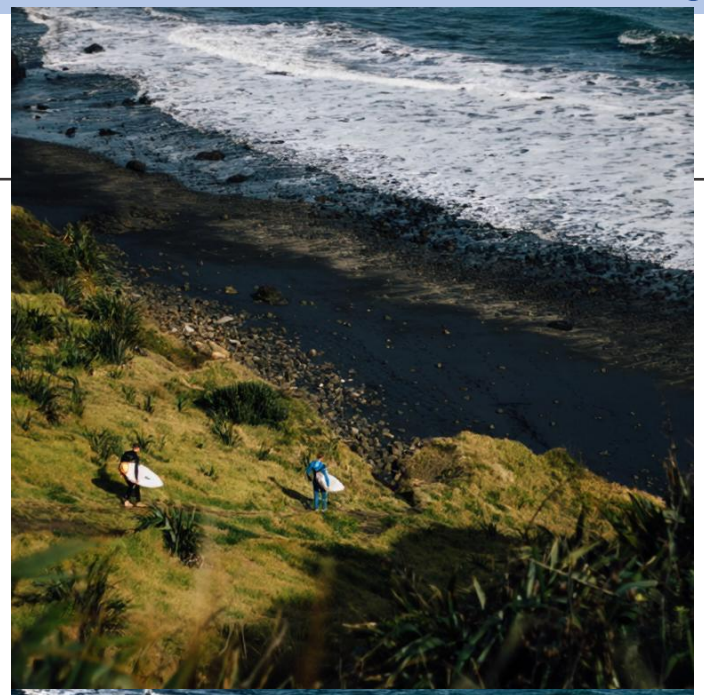




original

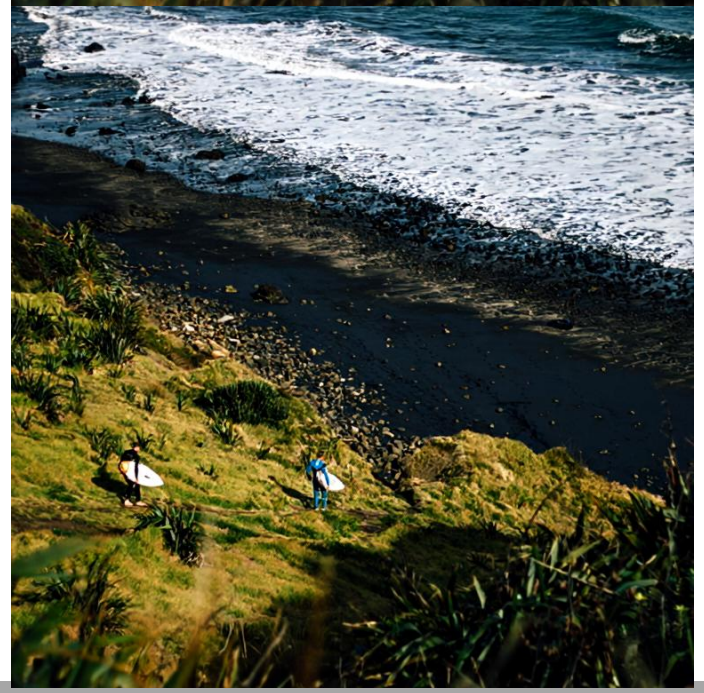


cs = 1.0



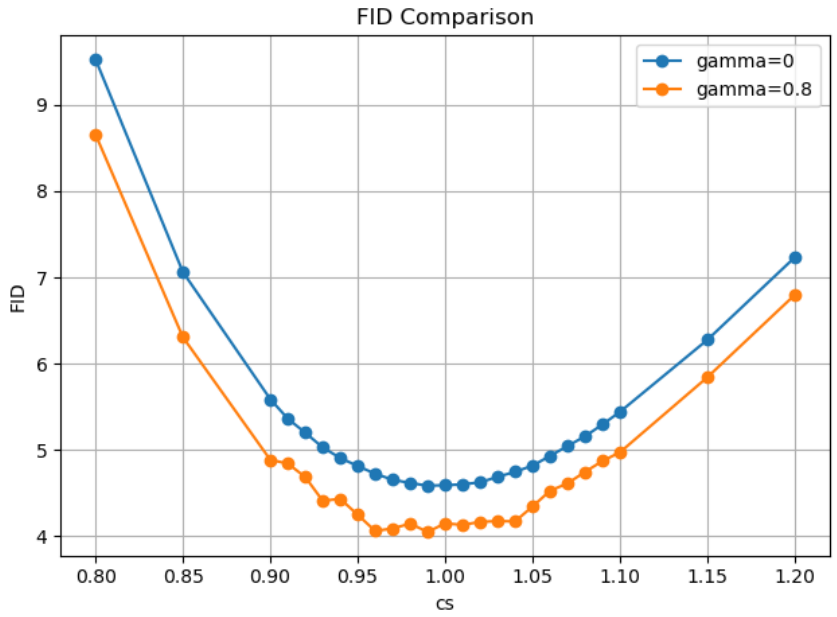
cs = 0.8

cs = 1.2

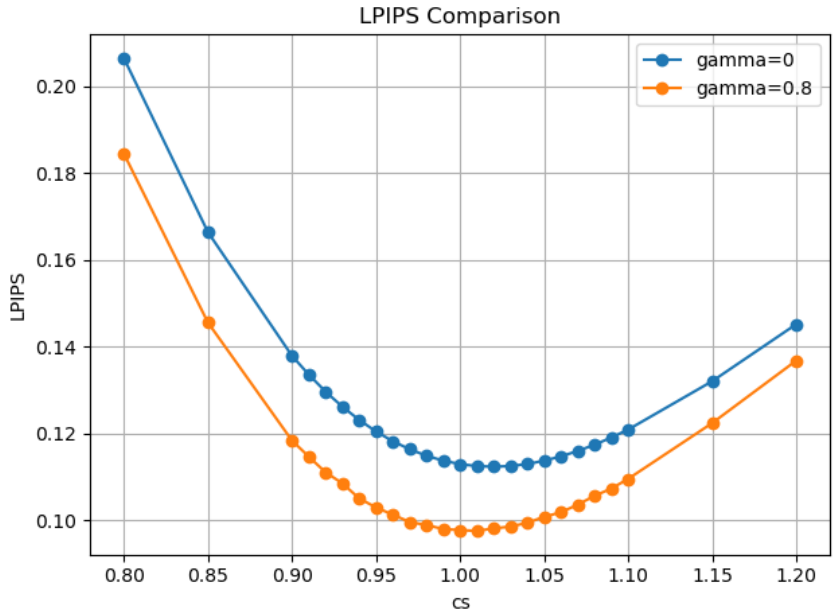


Evaluation result (compare γ = 0, 0.8)

- FID result
  - Results between cond_scale = 0.99 ~ 1.01 almost the same in γ = 0
  - Results between cond_scale = 0.96 ~ 0.99 are slightly better than scale = 1 in γ = 0.8
  - The image quality will decrease if distance between cond_scale and 1 increase



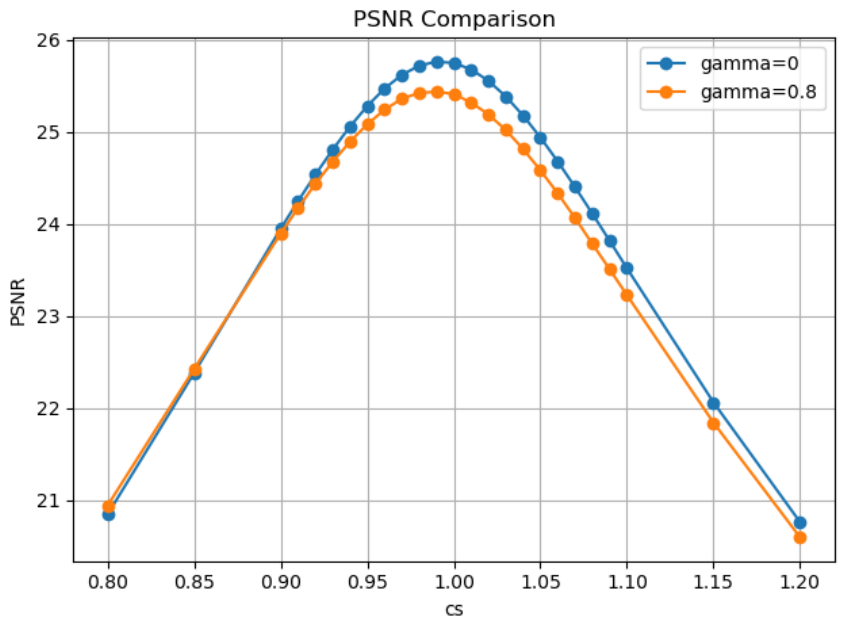
- LPIPS result
  - Results between cond_scale = 1.01 ~ 1.03 are slightly better than scale = 1 in γ = 0
  - Results between cond_scale = 0.99 ~ 1.01 almost the same in γ = 0.8
  - Show the same trend as FID result

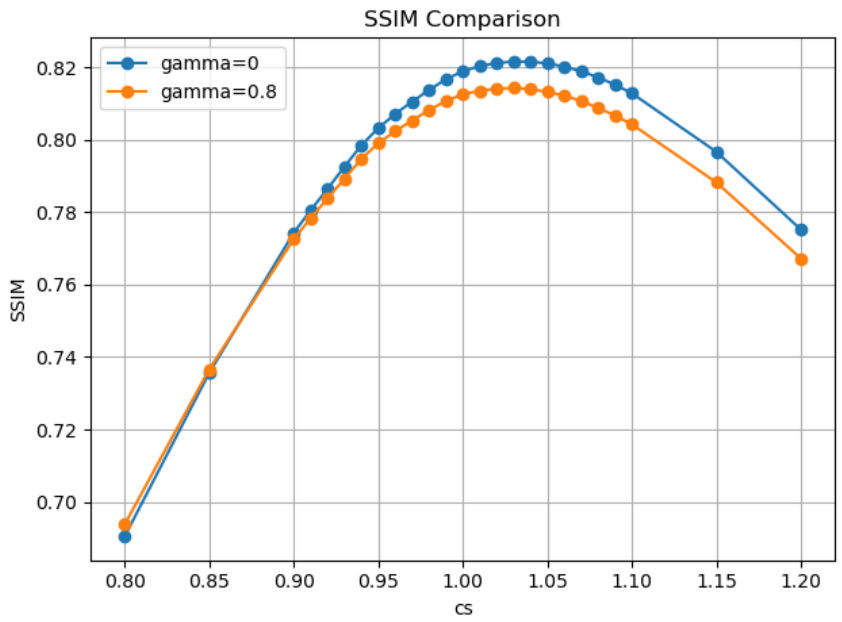


Evaluation result (compare γ = 0, 0.8)

- PSNR result
  - Results in cond_scale = 0.99 are slightly better than scale = 1 in γ = 0
  - Results between cond_scale = 0.99 ~ 1.01 almost the same in γ = 0.8
  - Show the same trend as FID result

- SSIM result
  - Results between cond_scale = 1.01 ~ 1.05 are slightly better than scale = 1 in γ = 0
  - Results between cond_scale = 1 ~ 1.03 almost the same in γ = 0.8
  - Show the same trend as FID result





Discussion

- Effect on compressed image
  - Amplifying the conditioning latent also amplifies the information passed to the decoder, which cause the model to **over-emphasize** structural and color cues
  - A weaker latent **reduces the decoder's confidence**, making it generate smoother, blurrier, and lower-contrast images

- Effect on metrics evaluation
  - cond_scale has a **global influence** on perceptual quality, not just contrast or brightness
  - The best perceptual performance is achieved when cond_scale stays **extremely close to 1**
  - **Small deviations** (0.96–1.04) may **slightly improve** quality, but larger deviations consistently degrade in every metrics
  - Reason: The decoder fails to provide the optimal solution since the **distribution shift too much** from training (if cond_scale's deviation is too large)

- Conclusion
  - Small deviations may slightly improve quality in some cases, but in general, cond_scale = 1 (as original structure) is still the best option.

Thank you.
Any questions?